

Road Racer Game Code guide

Step 1

As always we start with the code template. (A code template is a pre-made, fill-in-the-blank blueprint that lets programmers skip the setup and start coding faster.)

Inside our VS Code we type “!” and press enter. We should get code like this.

I already changed the title to say “Road Racer.”

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Road racer</title>
</head>
<body>

</body>
</html>
```

Step 2

Add style tags (after the title, just type “s” or “st” and you can choose style from the menu.)

I will add a body rule meaning I will style the entire body of the page, so it will have a green background color.

```
<title>Road racer</title>
<style>
  body{
    background-color: green;
  }
</style>
```

Now copy path and open in the browser. You should see a green screen.

Step 3

We now have a nice grassy background. Let's add a road.

Inside the body - between the <body> </body> tags - add a div.

What's a div?

Literally it means division, we want to group a few other elements like images, text, buttons etc inside the div. Think of it like a section of the website.

We give it an id of “road” so we can style it.

Heres the code and road style for now:

We use 210px width for 100px each lane and a 10px white line in the middle.

```
<title>Road racer</title>
<style>
  body{
    background-color: green;
  }
  #road{
    background-color: gray;
    width: 210px;
    height: 100vh;
  }
</style>
</head>
<body>
  <div id="road"></div>
</body>
</html>
```

We use “100vh”, which means 100% of the browser window’s height (called viewport height.) Using “100%” would only work if the parent element, such as the `body`, already had a set height. Since it does not, “100vh” is simpler and makes the road fill the screen.

And here's how it looks in the browser:



As you can see, there is a space between the road top and left. This is likely the automatic body margin.

We fix that by saying margin 0 in body:

```
body{
  background-color: green;
  margin: 0;
}
```

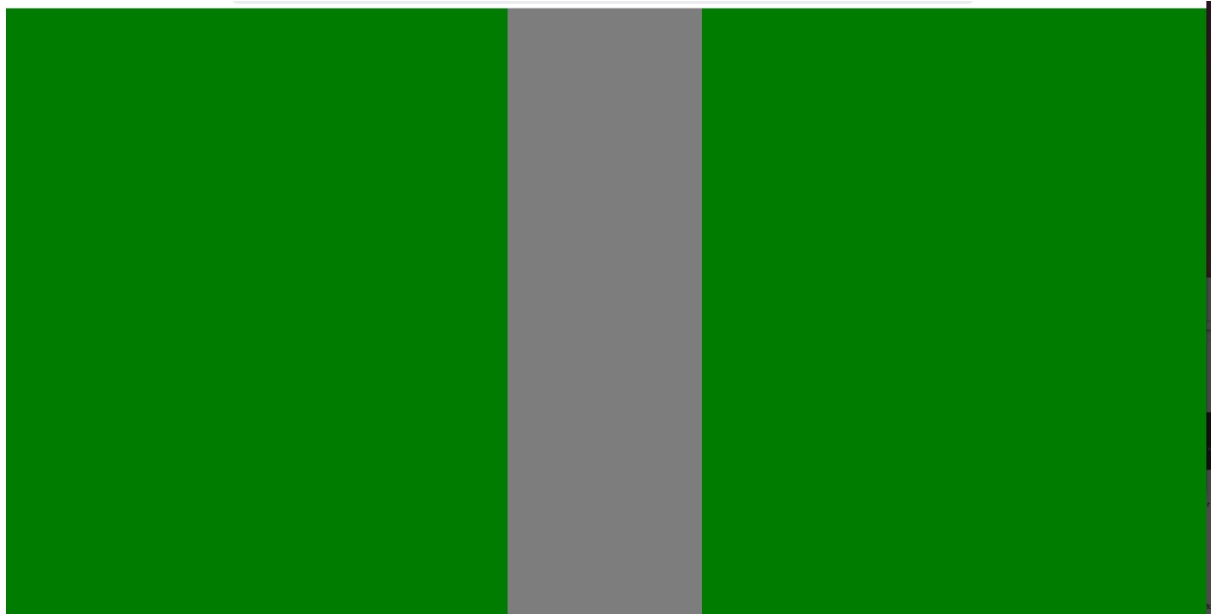
That should fix it!

To move the road to the middle, we use `margin: auto`. The browser puts the same amount of empty space on the left and right sides of the road.

We also use `position: relative`. The road stays in its normal place unless we give it `top`, `left`, `right`, or `bottom`. These would move the road relative to where it would normally be.

It also makes the road the starting point for absolutely positioned elements inside it. For example, a car with `left: 10px` will be placed 10 pixels from the left side of the road, not from the side of the screen.

```
#road {  
  background-color: gray;  
  width: 210px;  
  height: 100vh;  
  
  margin: auto;           /* Centers the road horizontally */  
  position: relative;     /* Makes the road the positioning area  
                           for absolute elements inside it */  
}
```



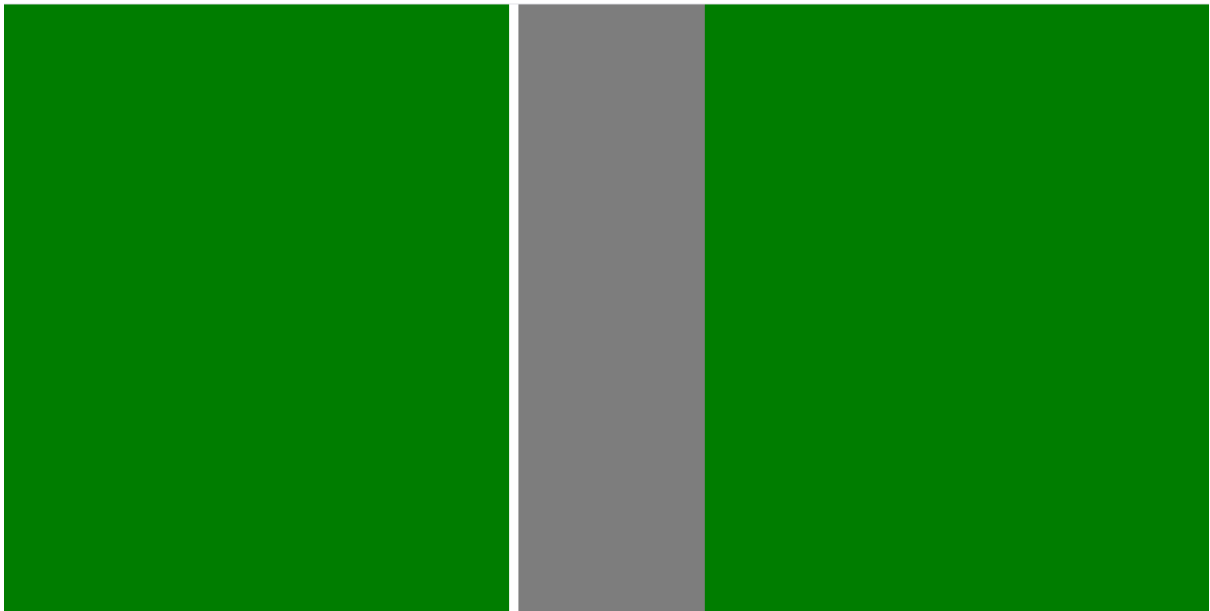
We now want a white line in the middle to look like a road - and to show 2 lanes.

We first place a div **inside** the road div.

```
<body>
  <div id="road">
    <div id="line"></div>
  </div>
</body>
```

In order to see our line div we must either add text (like this `<div id="line">white line</div>`), or style it. We want it white, same height as the parent or container div - the road, and 10px wide.

```
#line{
  width: 10px;
  height: 100%;
  background-color: white;
}
```



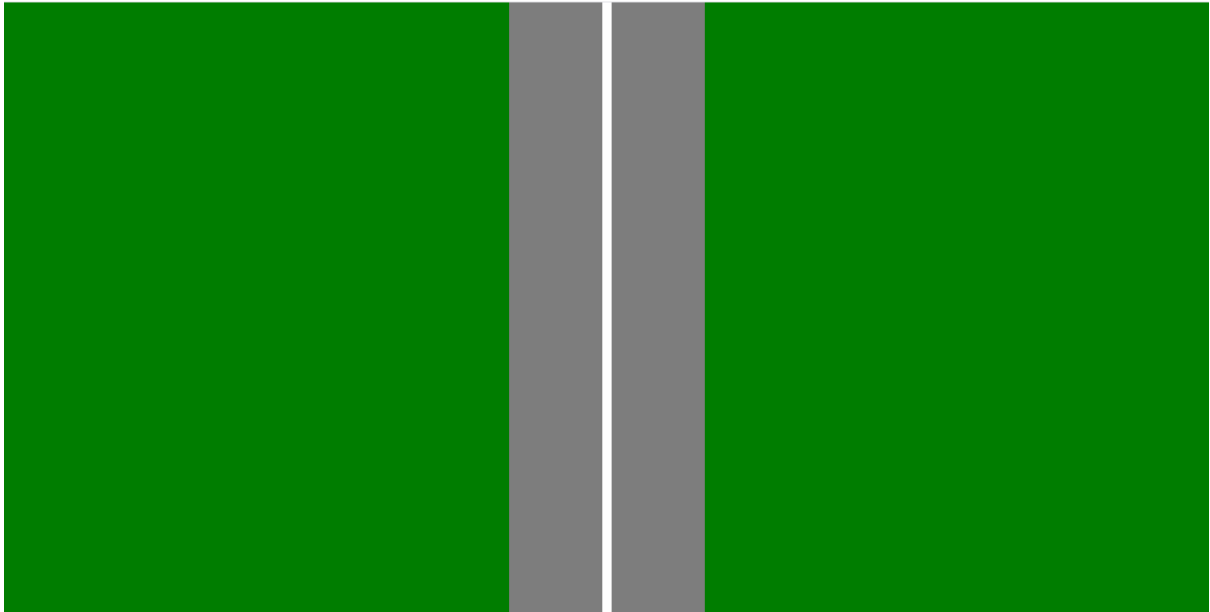
As you see, the line div automatically moves to the top left like any element.

How do we move it? We use absolute position.

But whats interesting this time is that since the container element is not the body like previously, but the road div, it uses that as its starting point for left, top etc.

```
#line{
  width: 10px;
  height: 100%;
  background-color: white;

  position: absolute;
  left: 100px;
}
```



We now add the player car image. This will also be nested inside the road element.
You can download the car images from here:

<https://rbscode.com/Student-Portal/images/37.png>

<https://rbscode.com/Student-Portal/images/38.png>

You can also simply use the link as your source.

```
<body>
  <div id="road">
    <div id="line"></div>

  </div>
</body>
```

Note, we give our image a class and an id. A class styles a few elements the same way, while an id only styles one element. We will have at least 2 cars, so we apply the car style to both with each getting its own styling according to its positioning.

We can place each attribute on a new line. This makes it easier to read.

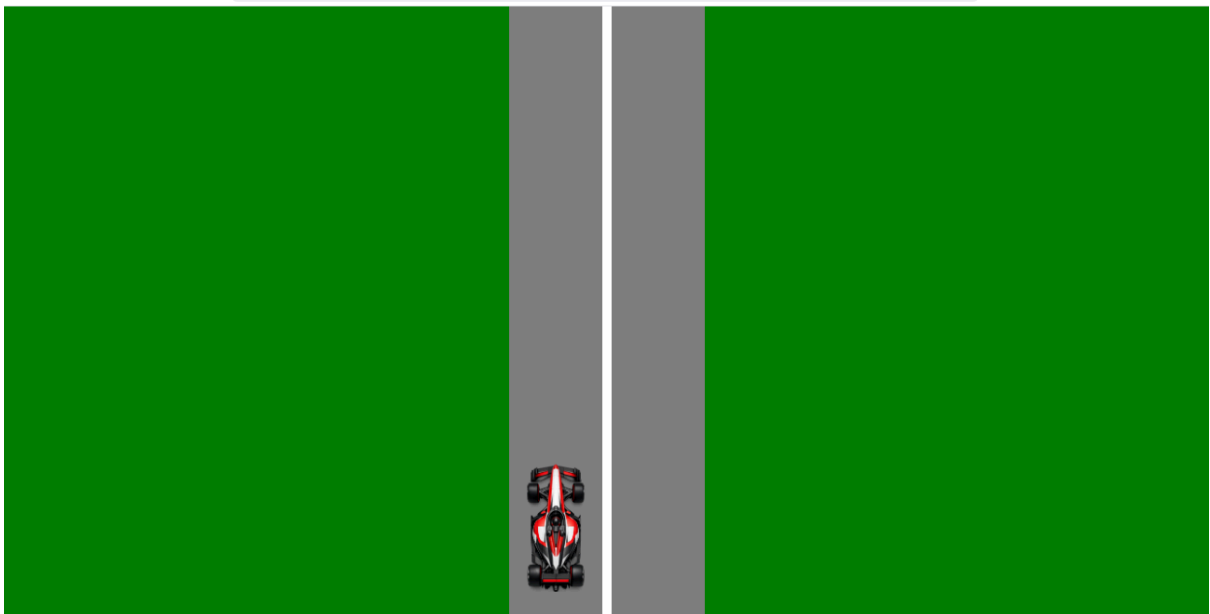
We can also keep it on one line:

```

```

Here's the car class and the player id styles:

```
/* Style all cars - using a class */
.car {
    width: 100px;
    position: absolute;
    /* border: 2px solid red; */
}
/* Style special player styles*/
#player{
    left: 0;
    bottom: 30px;
}
```



```
<body>
  <div id="road">
    <div id="line"></div>

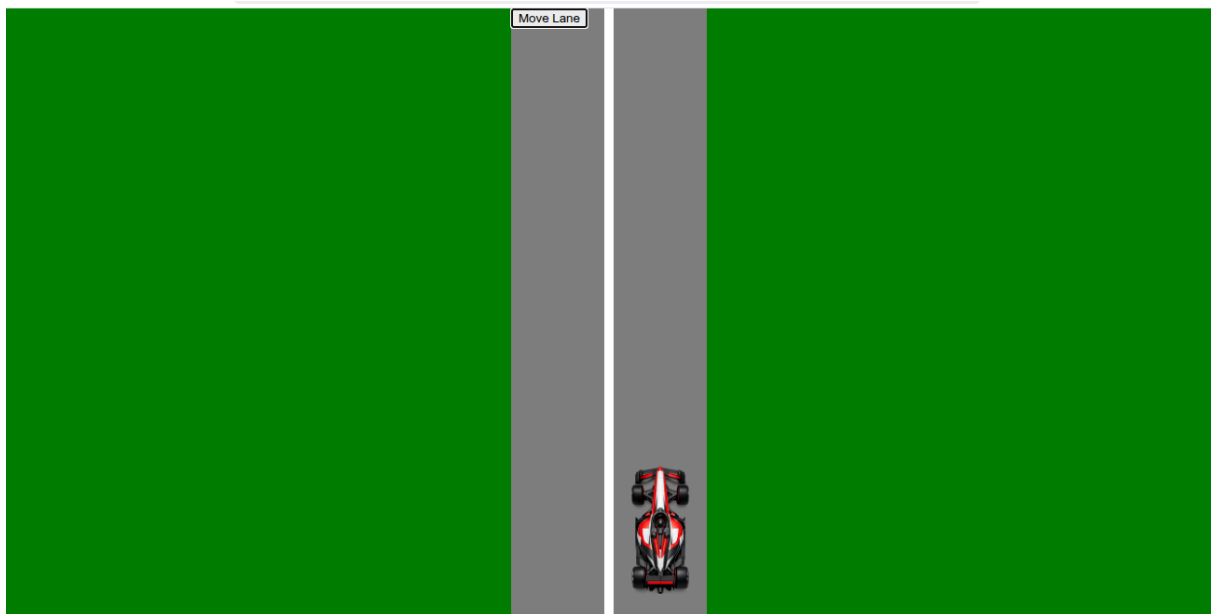
    

    <button onclick="moveLane()">Move Lane</button>

  </div>
```

Now lets create the move lane function.

```
<script>
  const player = document.getElementById("player")
  let playerLeft = 0
  function moveLane(){
    if(playerLeft == 0){
      playerLeft = 110
    }
    else{
      playerLeft = 0
    }
    player.style.left = playerLeft + "px"
  }
</script>
```



We can also make the car move using a key press.
Just add one more line:

```
<script>
  const player = document.getElementById("player")
  let playerLeft = 0
  document.addEventListener("keydown", moveLane)
  function moveLane() {
    if(playerLeft == 0){
      playerLeft = 110
    }
    else{
      playerLeft = 0
    }
    player.style.left = playerLeft + "px"
  }
</script>
```

We can now delete the button.

```
<body>
  <div id="road">
    <div id="line"></div>
    
  </div>

  <script>
```

Step

We now add a score box.

We will use it only later when we add a second car, but its good to see how the game will look and to understand how we add points.

Right at the top of our body code we add this div:

```
<body>

  <div id="score">
    Score: 0
  </div>
```

And style it like this just after the player:

```
#player{  
  left: 0;  
  bottom: 30px;  
}
```

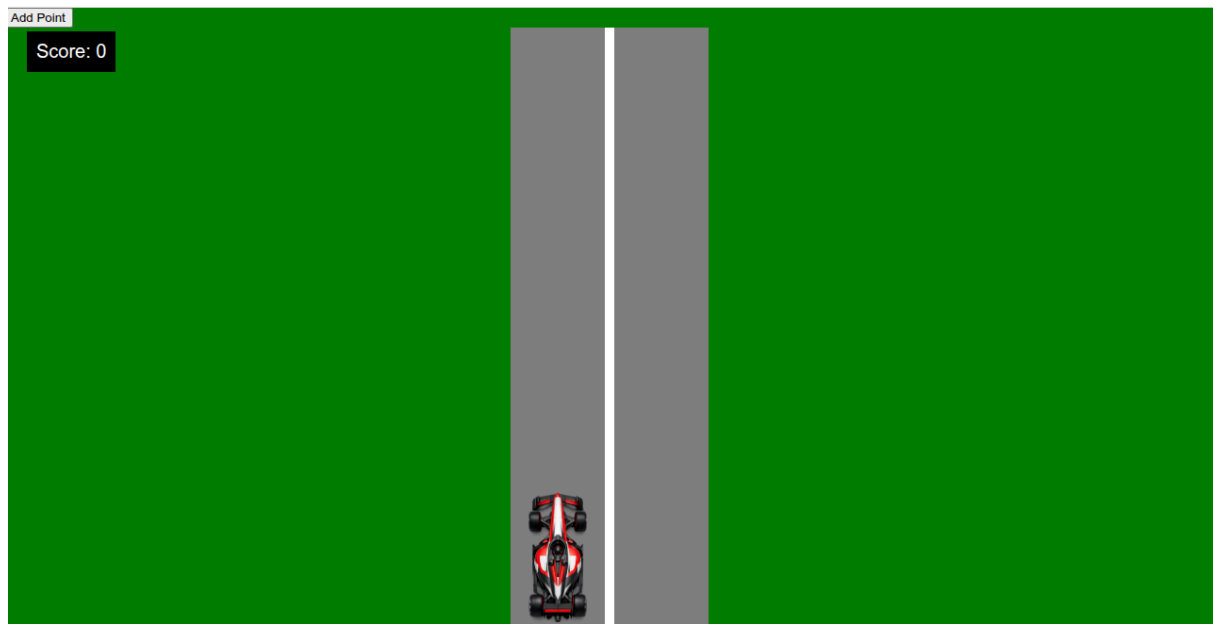
```
#score{  
  position: absolute;  
  top: 25px;  
  left: 25px;  
  
  padding: 10px;  
  
  color: white;  
  background: black;  
  font-family: Arial, Helvetica, sans-serif;  
  font-size: 20px;  
}  
</style>
```

Padding means the space between the text and the sides or edge of the button.

Now lets add a temporary button just before our score box to add points:

<body>

```
<button onclick="addPoint()">Add Point</button>  
  
<div id="score">  
  Score: 0  
</div>
```



The button pushes down the road a bit.

Dont worry, we can delete the button once we test the add point function and see the score go up.

We can also style the button with a position: absolute to take it out of the page flow.

And here's the add score function.

First we store our score box element in a variable. Remember a **const** is just a variable that is **constant** and cannot be changed.

```
<script>
  const player = document.getElementById("player")
  const scoreBox = document.getElementById("score")
  let score = 0

  function addPoint() {
    score++
    scoreBox.innerText = "Score: " + score
  }
```

Now, when you click the “Add Point” button, the score goes up.

We will delete this button soon.

Lets reorganize the code to look a bit neater.

Here's all the code so far. Please check to ensure you are doing ok:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width,
initial-scale=1.0">
<title>Road racer</title>
<style>
body{
    background-color: green;
    margin: 0;
}
#road {
    background-color: gray;
    width: 210px;
    height: 100vh;

    margin: auto;          /* Centers the road horizontally */
    position: relative;    /* Makes the road the positioning area
                           for absolute elements inside it */
}

#line{
    width: 10px;
    height: 100%;
    background-color: white;

    position: absolute;
    left: 100px;
}

/* Style all cars - using a class */
.car {
    width: 100px;
    position: absolute;
    /* border: 2px solid red; */
}

/* Style special player styles*/
#player{
    left: 0;
    bottom: 30px;
}

#score{
    position: absolute;
    top: 25px;
    left: 25px;
```

```

padding: 10px;

color: white;
background: black;
font-family: Arial, Helvetica, sans-serif;
font-size: 20px;
}
</style>
</head>
<body>

<button onclick="addPoint()">Add Point</button>

<div id="score">
  Score: 0
</div>

<div id="road">
  <div id="line"></div>
  
</div>

<script>
  // Get elements, store in variables
  const player = document.getElementById("player")
  const scoreBox = document.getElementById("score")

  // Game information.
  let playerLeft = 0
  let score = 0

  // let score = 0
  document.addEventListener("keydown", moveLane)
  function moveLane() {
    if(playerLeft == 0){
      playerLeft = 110
    }
  }

```

```

        else{
            playerLeft = 0
        }
        player.style.left = playerLeft + "px"
    }

    function addPoint() {
        score++
        scoreBox.innerText = "Score: " + score
    }
</script>
</body>
</html>

```

Step

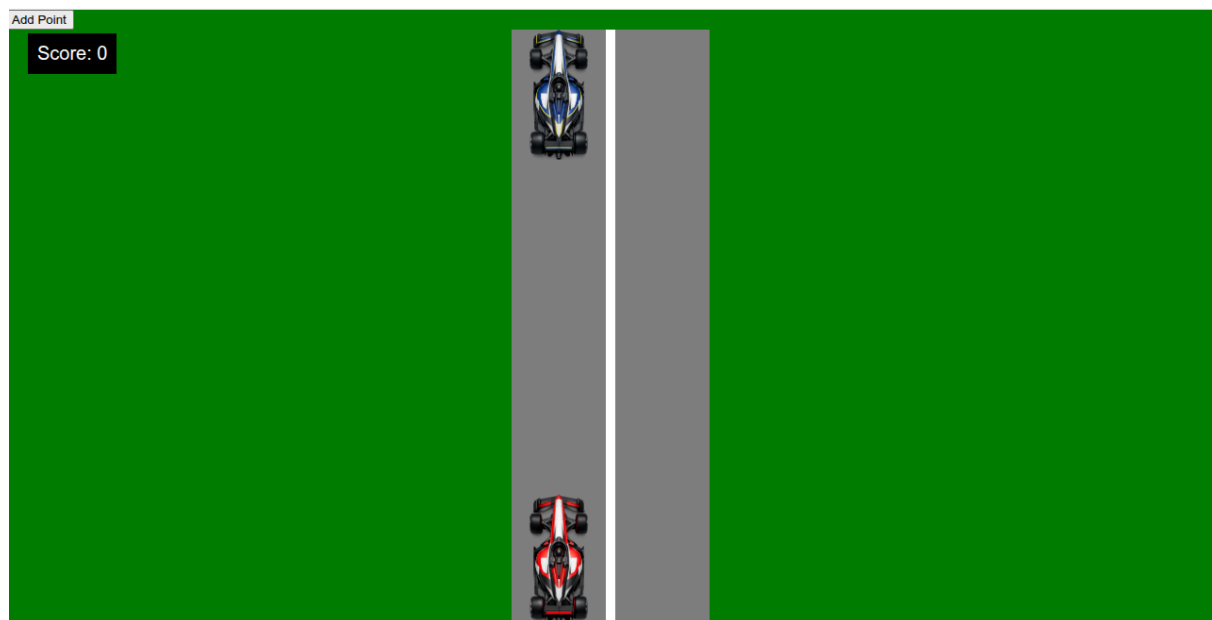
Adding enemy image

Place this right after player image inside the body:

```



```



Lets style it:

```
#enemy {  
  left: 0;  
  top: -160px;  
}
```

You can place it right after the #player styles. Note that the top is negative, meaning its

Now add another const for our enemy image element:

```
<script>  
  // Get elements, store in variables  
  const player = document.getElementById("player")  
  const enemy = document.getElementById("enemy") // New - enemy  
  const scoreBox = document.getElementById("score")  
  "score")
```

We add 3 variables for the enemy position and speed:

```
// Game information.  
let enemyTop = -160  
let speed = 12  
let enemyLeft = 0
```

And now for the move enemy function:

```
// Run the game again and again.  
  
function playGame() {  
  enemyTop += speed  
  enemy.style.top = enemyTop + "px"  
}  
game = setInterval(playGame, 20)
```

You will notice the car simply goes off the screen.

We need to detect when the car goes below the bottom edge.

```
// Run the game again and again.

function playGame() {
    enemyTop += speed
    enemy.style.top = enemyTop + "px"

    if (enemyTop > window.innerHeight) {
        enemyTop = -160
        addPoint() // delete temp addPoint score button
    }
}

game = setInterval(playGame, 20)
```